

Abdelhalim Azzouz

Professor of Mathematics

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Employment history

1999 – 2001	Part-time teacher, University of Tlemcen.
2001 – 2011	Assistant Professor, University of Saida, Algeria.
2011 – 2017	Associate Professor B, Department of Mathematics, Faculty of Science, University of Saida, Algeria.
2017 – 2023	Associate Professor A, Department of Computer Science, Faculty of Mathematics, Computer Science and Telecommunications, University of Saida, Algeria.
2023 – ..	Professor, Department of Computer Science, Faculty of Science, University of Saida, Algeria; then Department of Mathematics, Faculty of Sciences, University Salhi Ahmed, Naama, Algeria.

Education

1990	Baccalaureate, Technicum College, Tlemcen, Algeria.
1990 – 1996	Advanced diploma in Mathematical Analysis, University of Tlemcen, Algeria.
1999 – 2001	Magister, University of Tlemcen, Algeria. Thesis: <i>Optimization of flow rack AR/RS, mathematical tools for flexible manufacturing systems.</i>
2007 – 2011	Ph.D. in Mathematical Analysis (Functional Analysis), University of Oran 1, Algeria. Thesis: <i>On the sum and product of two closed operators in Hilbert space.</i>
2011 – 2018	Habilitation to Direct Research (HDR) in Mathematics, Department of Mathematics, Faculty of Science, University of Saida, Algeria.

Research publications

Journal articles

- [1] B. Abdellaoui, A. Azzouz, A. Bensedik, and R. Bentifour, “Fractional Kirchhoff–Hardy problem: breaking of resonance and general existence results,” *Axioms*, vol. 15, art. 199, 2026. [doi:10.3390/axioms15030199](https://doi.org/10.3390/axioms15030199)
- [2] A. Azzouz, M. Beghdadi, and B. Krichen, “On tauberian and co-tauberian spectra involving measures of weak noncompactness,” *Filomat*, 2026.
- [3] A. Azzouz, M. Beghdadi, and B. Krichen, “On left-right generalized Fredholm spectra and some perturbations results,” *Bull. Korean Math. Soc.*, vol. 62, no. 6, pp. 1767–1785, 2025. [doi:10.4134/BKMS.b240553](https://doi.org/10.4134/BKMS.b240553)
- [4] A. Azzouz, M. Beghdadi, and B. Krichen, “Some spectral properties of left and right generalized Fredholm operators,” *Georgian Math. J.*, vol. 32, no. 5, 2025. [doi:10.1515/gmj-2025-2004](https://doi.org/10.1515/gmj-2025-2004)
- [5] A. Azzouz, M. Beghdadi, and B. Krichen, “Correction to: g-Riesz operators and their spectral properties,” *Adv. Oper. Theory*, vol. 9, art. 88, 2024. [doi:10.1007/s43036-024-00385-y](https://doi.org/10.1007/s43036-024-00385-y)
- [6] H. Abbas, A. Azzouz, M. B. Zahaf, and M. Belmekki, “Generalized extended Riemann–Liouville

type fractional derivative operator,” *Kragujevac J. Math.*, vol. 47, no. 1, pp. 57–80, 2023. doi:[10.46793/KgJMat2301.057A](https://doi.org/10.46793/KgJMat2301.057A)

- [7] A. Azzouz, M. Beghdadi, and B. Krichen, “g-Riesz operators and their spectral properties,” *Adv. Oper. Theory*, vol. 7, no. 2, art. 18, 2022. doi:[10.1007/s43036-021-00179-6](https://doi.org/10.1007/s43036-021-00179-6)
- [8] A. Azzouz, M. Beghdadi, and B. Krichen, “Generalized relative essential spectra,” *Filomat*, vol. 36, no. 8, pp. 2657–2673, 2022. doi:[10.2298/FIL2208657A](https://doi.org/10.2298/FIL2208657A)
- [9] H. M. Dida, F. Hathout, and A. Azzouz, “On the geometry of the tangent bundle with vertical rescaled metric,” *Commun. Fac. Sci. Univ. Ank. Ser. A1 Math. Stat.*, vol. 68, no. 1, pp. 222–235, 2019. doi:[10.31801/cfsuasmas.443735](https://doi.org/10.31801/cfsuasmas.443735)
- [10] G. Djellouli, A. Azzouz, and B. Messirdi, “Essential spectra and Drazin invertibility of quasisimilar closed operators,” *An. Științ. Univ. Al. I. Cuza Iași Mat. (N.S.)*, vol. 64, no. 2, pp. 221–231, 2018.
- [11] A. Azzouz, “A review on weak similarities relations of bounded operators,” *Adv. Stud. Contemp. Math.*, vol. 26, no. 4, pp. 607–626, 2016.
- [12] A. Azzouz, B. Messirdi, and G. Djellouli, “New results on the closedness of the product and sum of closed linear operators,” *Bull. Math. Anal. Appl.*, vol. 3, no. 2, pp. 151–158, 2011.
- [13] B. Messirdi, M. H. Mortad, A. Azzouz, and G. Djellouli, “A topological characterization of the product of two closed operators,” *Colloq. Math.*, vol. 112, no. 2, pp. 269–278, 2008.

Preprints

- [1] A. Azzouz, P. Agarwal, D. F. M. Torres, and H. Abbas, “Towards a Gagliardo-type theory of fractional Sobolev spaces on arbitrary time scales,” [arXiv:2603.14715](https://arxiv.org/abs/2603.14715), 2026.
- [2] H. Abbas and A. Azzouz, “A nonlocal transmission problem on a hybrid continuous-discrete domain,” [arXiv:2603.14178](https://arxiv.org/abs/2603.14178), 2026.
- [3] B. Abdellaoui, A. Azzouz, and A. Bensedik, “Fractional Kirchhoff problem with Hardy potential: breaking of resonance,” [arXiv:1811.09268](https://arxiv.org/abs/1811.09268), 2018.

Communications and schools

- CIMPA–UNESCO school, Control of PDE, Tlemcen (Algeria), 26 April – 8 May 2003.
- Winter School of Geometry and Cosmology, University of Mascara, 17–22 December 2005.
- School of geometry and dynamical systems, El Marsa, Algiers, 6–13 April 2006.
- Symposium *Équations différentielles et Géométrie différentielle* (organizer), Saida, 11–14 November 2007.
- *On almost closable operators*, École du printemps d’analyse fonctionnelle, Rabat (Morocco), 18–23 May 2009.
- *On triviality of sum and product of closed operators*, ICOT, Kerkennah (Tunisia), 2012.
- *On sum and product of closed operators*, International Conference on Advances in Applied Mathematics, Hammamet (Tunisia), December 2014.
- CIMPA school *Nonlinear partial differential equations arising from geometry and physics*, Hammamet (Tunisia), March 2015.
- CIMPA–CIRM school *Nouvelles perspectives sur les équations d’évolution non linéaires*, CIRM, Marseille (France), 3–9 April 2016.
- *On the geometry of the tangent bundle with vertical rescaled metric*, 14th International Geometry Symposium, Denizli (Turkey), 25–28 May 2016.
- *Further extensions of some Eulerian functions*, ICAAM, Djerba (Tunisia), 2022.
- *Generalized extended Riemann–Liouville fractional derivative*, ICMSA, Guelma (Algeria), 2023.

Teaching

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- Basic mathematical analysis.
 - Linear and bilinear algebra.
 - Unconstrained and constrained optimization.
 - Operator theory, spectral theory.
 - Numerical analysis for engineers.

This list is not exhaustive. As a mathematics lecturer for more than two decades, I have taught core and advanced courses, together with tutorials, to students in the Mathematics, Computer Science, Process Engineering and Commercial Sciences tracks.

Books

Published

- A. Azzouz, *Fonctions hypergéométriques*, Éditions universitaires, 2023.
- A. Azzouz, *Théorie des opérateurs (Étude spectrale)*, Éditions universitaires, 2023.

Forthcoming

- A. Azzouz and K. Djerfi, *Au-delà de Riemann : Mesure, intégration et passage à la limite*, Édition SarahMed (Licence 3 course with exercises, Master complements).
- K. Djerfi and A. Azzouz, *Calcul différentiel dans $\mathbb{R}^n$* , Office des Publications Universitaires (OPU).

Supervision

Ph.D. thesis

- M. Beghdadi (advisor B. Krichen, co-advisor A. Azzouz), *Development of some spectral properties involving the class of tauberian operators*, defended at Sfax (Tunisia), 13 November 2021.

Master theses (University of Saida)

- Allou Zerga, *S-essential spectra of bounded operators*, 25 June 2018.
- Nouari Aicha, *Convergence of operators*, 25 June 2018.
- Chibani Samira, *Confluent functions*, 16 July 2019.
- Ziani Hichem, *Spectral continuity of sequence of bounded operators*, 16 July 2019.
- Aliouane Amine, *Fractional Laplacian*, 16 July 2019.

Scientific visits

10–25 June 2025	International Telematic University UNINETTUNO, Rome, Italy. Invited by Pr. Clemente Cesarano.
10–25 May 2024	International Telematic University UNINETTUNO, Rome, Italy. Invited by Pr. Clemente Cesarano.
10–25 January 2017	Laboratoire de géométrie, analyse et applications, University of Paris 13. Invited by Pr. H. Zaag, PDE research group.
10–20 March 2015	Laboratoire de Mathématique-Physique, Faculty of Sciences, Sfax. Invited by Pr. A. Jeribi, research group on operator theory and fixed point theory.
10–20 July 2014	Laboratoire d'analyse mathématique, Seville, Spain. Invited by Pr. A. M. Rodríguez, operator theory research group.
10–20 March 2012	Laboratoire de Mathématique-Physique, Faculty of Sciences, Sfax. Invited by Pr. A. Jeribi, research group on operator theory and fixed point theory.

10–20 March 2010

Laboratoire de Mathématique-Physique, Faculty of Sciences, Sfax. Invited by Pr. A. Jeribi, research group on operator theory and fixed point theory.

Skills

Languages	Strong reading, writing and speaking competence in English and French.
Coding	Matlab, Python, $\text{\LaTeX}$.
Miscellaneous	Academic research, public animation, training, consultation, $\text{\LaTeX}$ typesetting and publishing.